

Section: Electricity

Topic: Capacitors

This question involves a capacitor discharging in a circuit.

(a)
Calculate the energy stored in the capacitor before discharge

Use the energy formula:

$$E = \frac{1}{2} CV^2$$

Given:

- $C = 680 \times 10^{-6} \text{ F}$
- $V = 12 \text{ V}$

$$E = \frac{1}{2} \times 680 \times 10^{-6} \times 12^2 = 0.5 \times 680 \times 10^{-6} \times 144 = 0.049 \text{ J}$$

(b)
Calculate the charge on the capacitor after 2.0 seconds

Use the exponential decay equation for charge:

$$Q = Q_0 e^{-t/RC}$$

Given:

- $Q_0 = CV = 680 \times 10^{-6} \times 12 = 8.16 \times 10^{-3} \text{ C}$
- $t = 2.0 \text{ s}, R = 470 \, \Omega, C = 680 \times 10^{-6} \text{ F}$

Calculate time constant:

$$RC = 470 \times 680 \times 10^{-6} = 0.32 \text{ s}$$

$$Q = 8.16 \times 10^{-3} \times e^{-2.0/0.32} = 8.16 \times 10^{-3} \times e^{-6.25} \approx 8.16 \times 10^{-3} \times 0.00193 = 1.6 \times 10^{-5} \text{ C}$$

 Revision Tips

- Use $Q = Q_0 e^{-t/RC}$ for exponential discharge
- Time constant $\tau = RC$ tells you how fast discharge occurs
- Energy stored in capacitor is proportional to **square of voltage**:

$$E = \frac{1}{2} CV^2$$

- Check units: capacitance in **farads**, time in **seconds**